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Assignment No: 01

Title of Assignment: Study different types of Attacks as a part of awareness: Phishing,
Ransomware and Supply-Chain attack.

Date of Performance: 14/07/2025

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Assignment 1: Study different types of Attacks as a part of awareness: Phishing, Ransomware and Supply-Chain attack.

Objective:

To understand and analyze various types of cyber-attacks: Phishing, Ransomware, and Supply Chain Attacks and to create awareness about how these attacks work, their impact, and prevention techniques.

Materials Required:

Computer with internet access

Introduction:

1. Phishing

Phishing is a **cyberattack** method that uses **social engineering techniques** to deceive users into providing **confidential information** such as usernames, passwords, credit card numbers, or personal details. Attackers often impersonate legitimate institutions (banks, government bodies, or popular platforms like Google or Amazon) to gain the victim's trust.

Key Characteristics of Phishing:

- **Deceptive communication** (email, SMS, calls)
- Use of **fake websites or login pages**
- Exploits **human psychology** (urgency, fear, curiosity)
- Aims to steal sensitive data or install malware

Types of Phishing

- **Email Phishing:** Broadly distributed emails mimicking reputable organizations, like banks or retailers, enticing users to click harmful links or share confidential data.
- **Spear Phishing:** Customized attacks targeting specific individuals or companies, using tailored details (e.g., from LinkedIn or corporate sites) to enhance authenticity.
- **Smishing:** Phishing through text messages, delivering malicious links or prompts for sensitive information.
- **Vishing:** Phone-based phishing where attackers impersonate credible sources, such as government officials or company representatives, to extract data.

Example of a Phishing Email:

Subject: Urgent Security Alert – Verify Your Account Now

From: support@bankofamerica-secure.com

Dear Customer,

We noticed suspicious activity on your account. Please verify your identity immediately to prevent suspension.

Click here to secure your account: <http://bank-verify-login.com>

Regards,

Bank of America Security Team

Phishing Indicators:

- **Generic greeting** (e.g., "Dear Customer" instead of your name)
- **Urgent language** (e.g., "immediate action required")
- **Suspicious links** (hover to reveal mismatched URLs)
- **Poor grammar or spelling mistakes**
- **Unexpected attachments**
- **Sender address mismatch** (e.g., support@bankofamerica-secure.com is not official)

Real-World Examples of Phishing Attacks:

- Google & Facebook Scam (2013–2015):

- **Attacker:** Evaldas Rimasauskas
- **Method:** Fake invoices and emails pretending to be from Quanta (a vendor)
- **Impact:** \$100+ million stolen from Google and Facebook



(Figure 1a: Money phishing with messenger)

Page at Risk

To: [REDACTED]



Hello,

We have received several reports against your activity!
You have posted many videos that violate our rules.
Reports claimed that your posts violated Facebook Policies regarding third party intellectual property rights.
Based on our policies, robotic growth of followers , sensitive content, human rights violators, animal rights violators or even copyrights should be prohibited.
Referring to reports on your page, your page has been set up in the deletion process.

Please first look at the activity on your site and if you think this is a mistake , please contact us.
For appeal and for more details below at the link.

<https://www.facebook.com/104119724874987/>

Appeals after 48 hours won't be taken into consideration

Sincerely,
Facebook Team

(Figure 1b: Facebook phishing scam)

Prevention & Protection Tips:

1. **Be cautious** of unsolicited emails and messages.
2. **Do not click** on suspicious links or download unexpected attachments.
3. **Verify the sender's email address.**
4. **Use 2-Factor Authentication (2FA)** wherever possible.
5. **Install anti-phishing toolbars** and **browser protection extensions.**
6. **Educate users** through cybersecurity awareness programs.
7. **Regularly update** your systems and software.

2. Ransomware

Ransomware is a type of malicious software (malware) that **encrypts files** on a victim's device, making them inaccessible. The attacker then demands a **ransom**, usually in cryptocurrency (like Bitcoin), in exchange for a decryption key.

Key Characteristics of Ransomware:

- Encrypts important files or entire systems.
- Displays ransom note demanding payment.
- Often spreads via phishing emails, malicious downloads, or software vulnerabilities.
- May also steal data before encryption (double extortion).

Types of Ransomware:

- **Crypto Ransomware:** Encrypts files (e.g., *WannaCry*, *Maze*).

- **Locker Ransomware:** Locks access to the device without encrypting data.
- **Scareware:** Fake alerts urging payment for fake threats.
- **Doxware:** Threatens to leak stolen data publicly if ransom isn't paid.

Real-World Examples of Ransomware Attacks:

- **WannaCry (2017):**
 - **Spread:** Exploited a vulnerability in Windows (EternalBlue).
 - **Impact:** Affected over 200,000 computers in 150 countries including UK's NHS.
 - **Ransom:** Demanded Bitcoin for decryption.
- **NotPetya (2017):**
 - **Disguised as:** A software update from a Ukrainian company.
 - **Impact:** Hit global companies like Maersk, Merck, FedEx.
 - **Damage:** Estimated over \$10 billion in losses.



(Figure 2a: Cryptolocker)

CryptoLocker, an encrypting Trojan horse, occurred from 5 September 2013 to late May 2014.

The Trojan targeted computers running Microsoft Windows, propagating via infected email attachments and via an existing Gameover Zeus botnet.

Once activated, the malware encrypted files stored on local and mounted network drives using RSA public-key cryptography, with the decryption key stored on the malware's control servers.

CryptoLocker then displayed a ransom message offering to decrypt the data if a Bitcoin or prepaid cash voucher payment was made by a stated deadline. It employed social engineering to create a sense of urgency, threatening to delete the decryption key if the deadline passed.

How It Works:

1. Ransomware is delivered via email or infected websites.
2. It encrypts files and locks the system.
3. A ransom note is displayed demanding payment for decryption.

Prevention & Protection Tips:

1. **Back up data regularly** (offline and in the cloud).
2. Use **updated antivirus software** and enable real-time protection.
3. **Patch vulnerabilities** with regular software updates.
4. Don't open **unsolicited attachments or links** in emails.
5. **Disable macros** in documents from unknown sources.
6. Implement **Network Segmentation** to limit malware spread.
7. Use **Application Whitelisting** to prevent unauthorized software.

3. Supply Chain Attack

Supply Chain Attacks occur when cybercriminals **target trusted third-party vendors** or software providers to infiltrate larger organizations. Instead of attacking directly, they exploit vulnerabilities in a partner's infrastructure, software, or update mechanisms.

Key Characteristics of Supply Chain Attacks:

- Compromise trusted software/code providers.
- Malicious code is inserted into updates, plugins, or hardware components.
- Difficult to detect due to trusted nature of source.
- Allows **widespread access** to many downstream victims.

Types of Supply Chain Attacks:

- **Software Supply Chain Attack:** Compromising software updates (e.g., SolarWinds).
- **Hardware Supply Chain Attack:** Inserting backdoors in hardware components.
- **Third-Party Vendor Attack:** Using insecure vendors or cloud services to gain access.

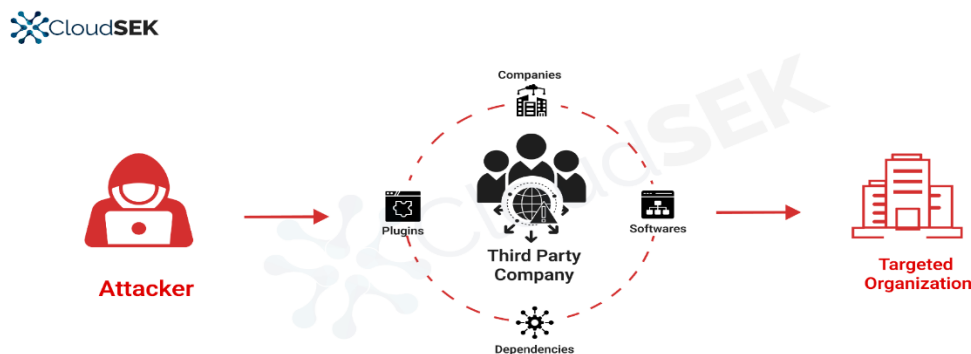
Real-World Example: SolarWinds Attack (2020)

- **Targeted Software:** Orion IT monitoring platform by SolarWinds.
- **What Happened:** Hackers inserted malware (**SUNBURST**) in software updates.
- **Victims:** Over 18,000 organizations including U.S. Treasury, Microsoft, and FireEye.

- **Attribution:** Believed to be state-sponsored (APT29/Russian group).
- **Impact:** Massive data breaches, loss of trust, national security concerns.



(Figure 3a: Supply chain attack)



(Figure 3b: CloudSEK Monitoring supply chain)

Prevention & Protection Tips:

1. Conduct **due diligence on vendors** and their security practices.
2. Use **code-signing certificates** for software authenticity verification.
3. Implement **Zero Trust Architecture** (trust nothing, verify everything).
4. Monitor **network traffic and system logs** for anomalies.
5. Enforce **least privilege access** for third-party integrations.
6. Regularly **audit and review supply chain dependencies**.
7. Use **endpoint detection and response (EDR)** tools.

Conclusion: Understanding cyberattacks like phishing, ransomware, and supply chain attacks is crucial in today's digital world. These threats exploit both technical vulnerabilities

and human behavior, making awareness and proactive security measures essential. Regular training, system updates, and strong security practices can significantly reduce the risk of falling victim to such attacks.